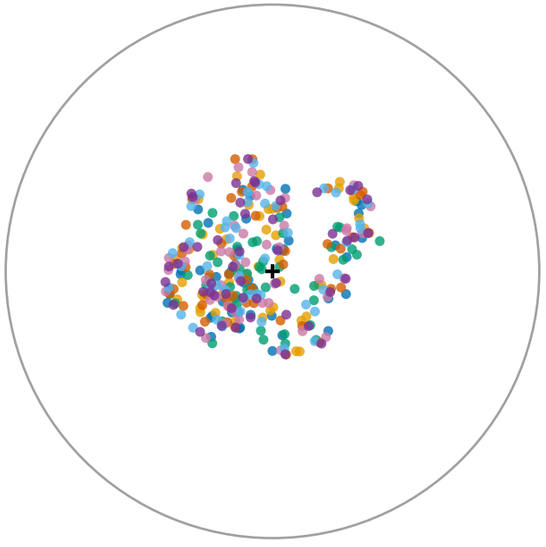
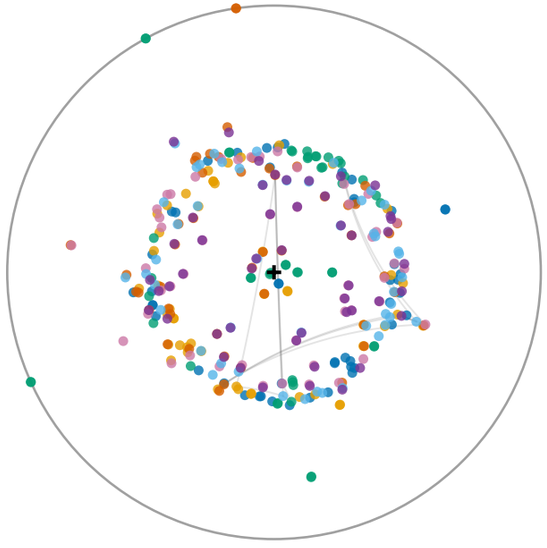
MSRVTT · hyperbolic + cone · lambda = 0.1 · training seed 50

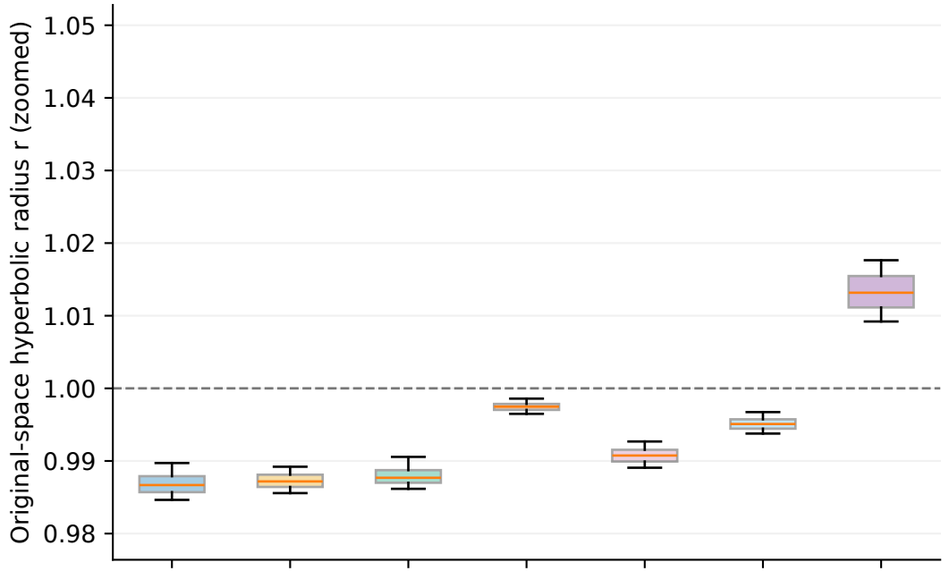
8D · HoroPCA · projected radii are distorted
distance ρ=0.71; kNN10=40.1%



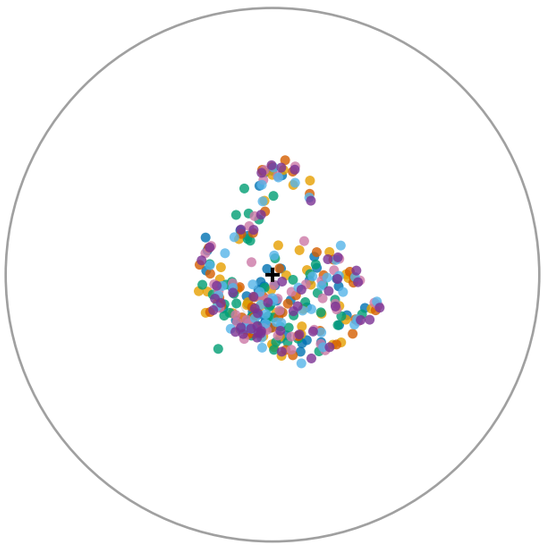
8D · CO-SNE · projected radii are distorted
distance ρ=0.08; kNN10=23.5%



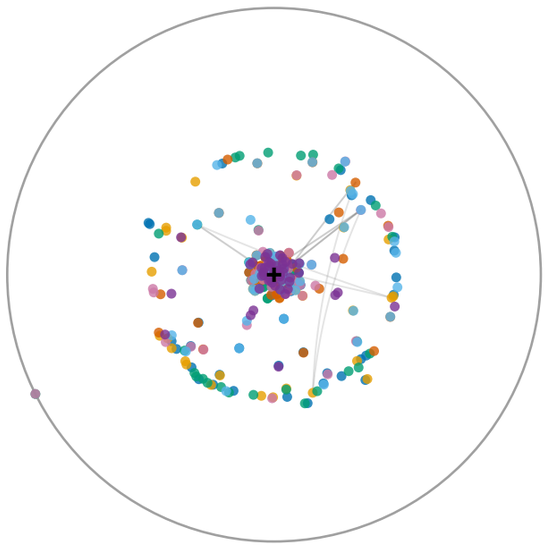
8D · actual radius · best epoch 7
Strict cone satisfaction: 0.00%



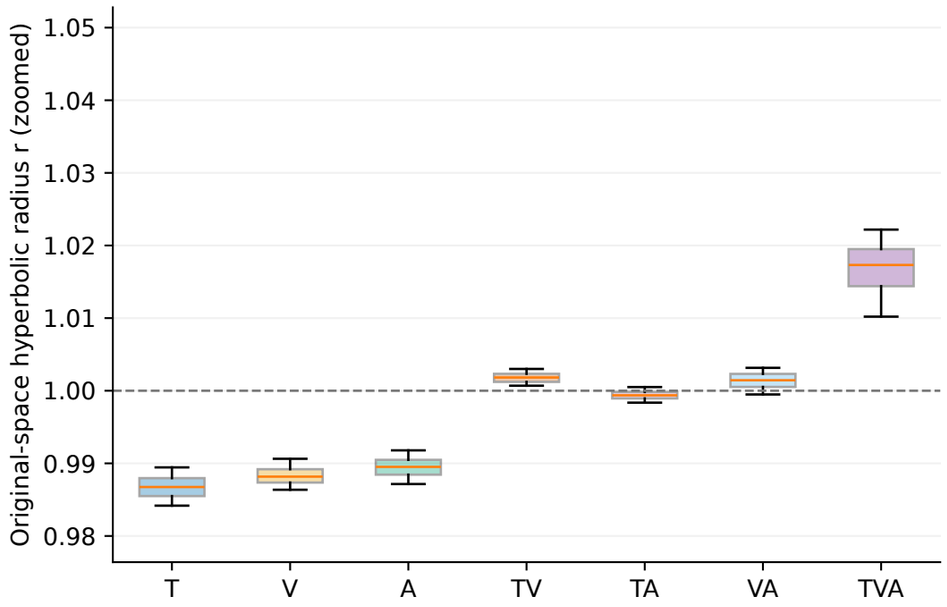
16D · HoroPCA · projected radii are distorted
distance ρ=0.53; kNN10=29.9%



16D · CO-SNE · projected radii are distorted
distance ρ=0.02; kNN10=16.9%



16D · actual radius · best epoch 7
Strict cone satisfaction: 0.00%

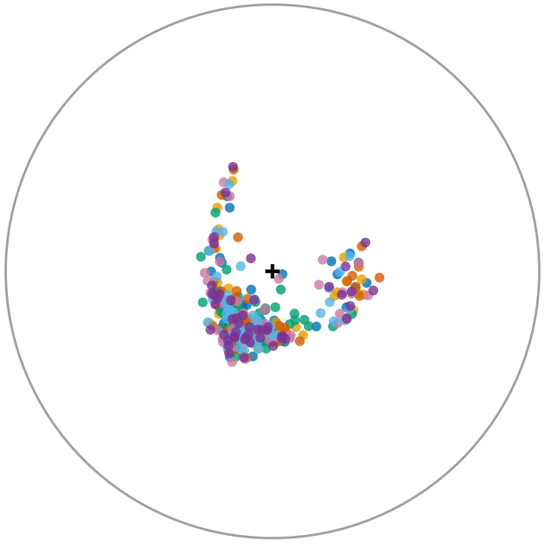


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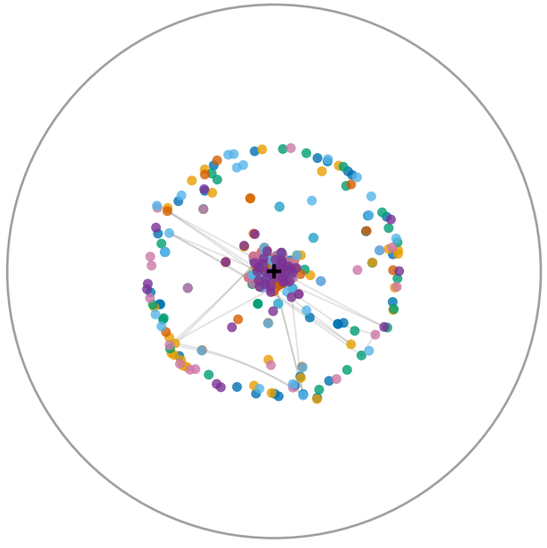
Same 48 test videos × 7 subsets; projection seed 50 only. Right column: all 884 videos. Gray lines are prescribed edges, not confirmed containment.

MSRVTT · hyperbolic + cone · lambda = 0.1 · training seed 50

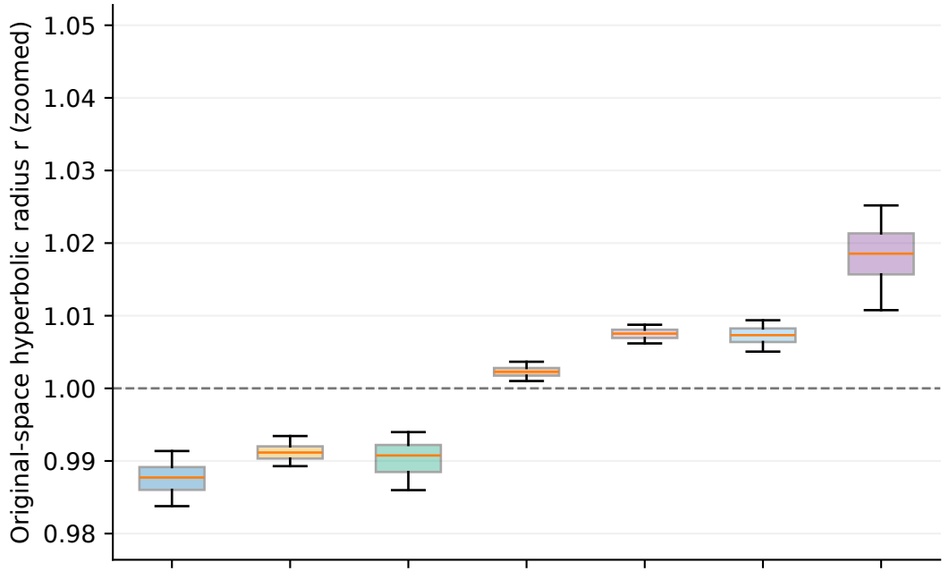
32D · HoroPCA · projected radii are distorted
distance ρ=0.35; kNN10=26.5%



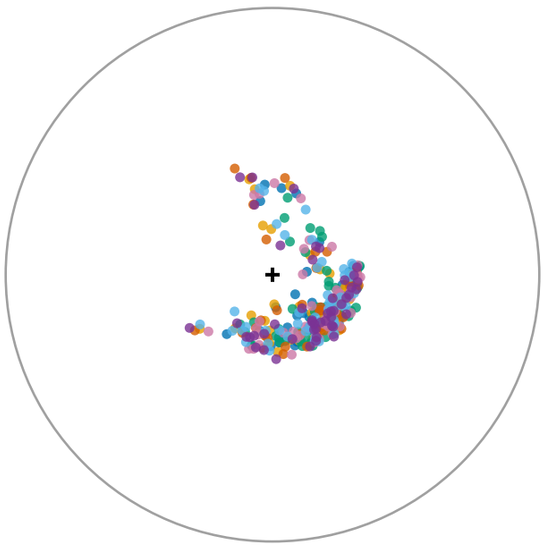
32D · CO-SNE · projected radii are distorted
distance ρ=0.05; kNN10=16.6%



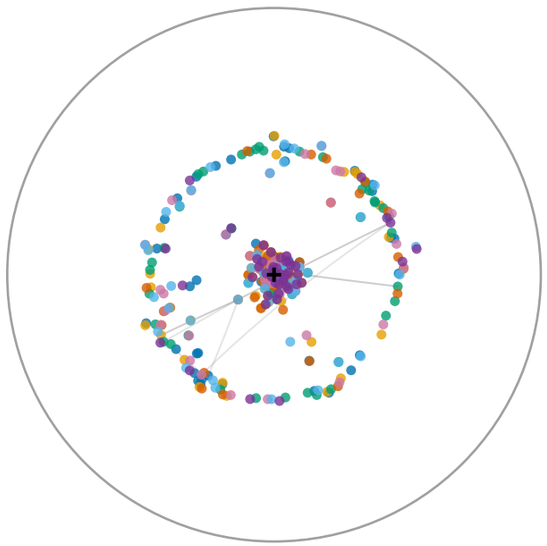
32D · actual radius · best epoch 8
Strict cone satisfaction: 0.00%



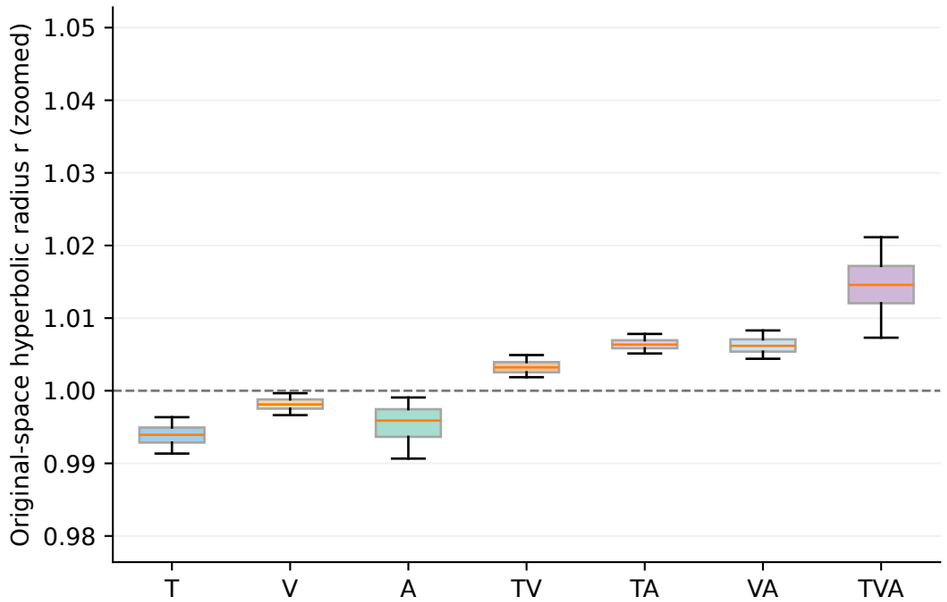
64D · HoroPCA · projected radii are distorted
distance ρ=0.36; kNN10=27.7%



64D · CO-SNE · projected radii are distorted
distance ρ=0.01; kNN10=14.0%



64D · actual radius · best epoch 5
Strict cone satisfaction: 0.00%

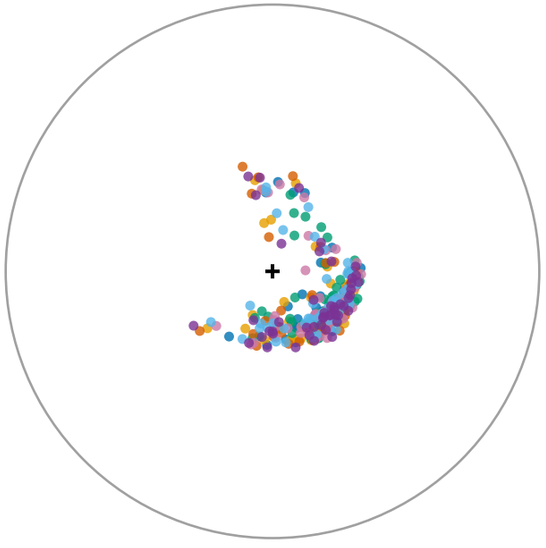


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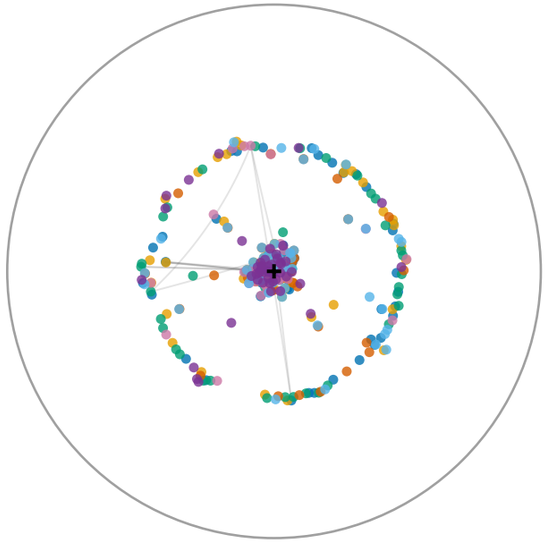
Same 48 test videos × 7 subsets; projection seed 50 only. Right column: all 884 videos. Gray lines are prescribed edges, not confirmed containment.

MSRVTT · hyperbolic + cone · lambda = 0.1 · training seed 50

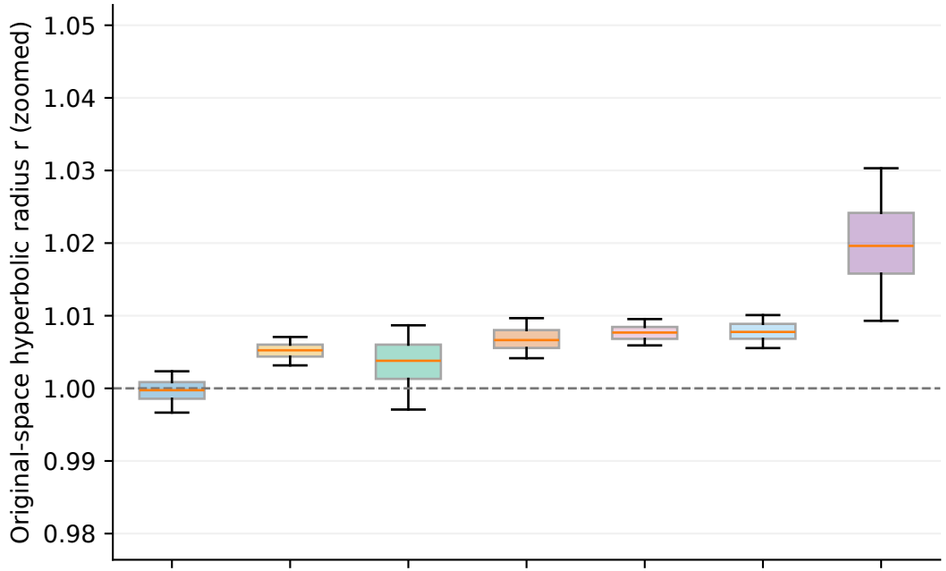
128D · HoroPCA · projected radii are distorted
distance ρ=0.35; kNN10=27.8%



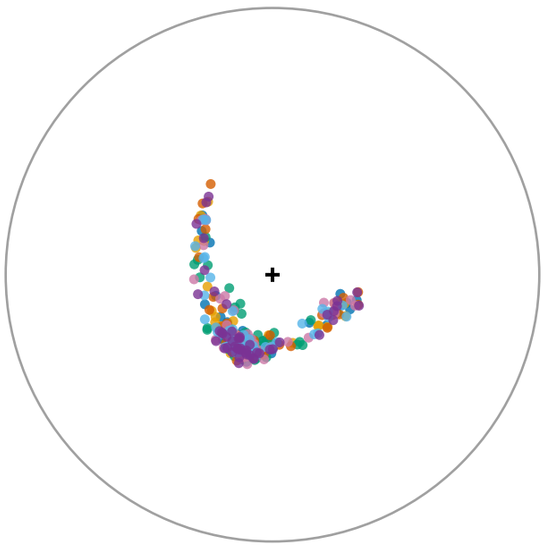
128D · CO-SNE · projected radii are distorted
distance ρ=0.01; kNN10=15.2%



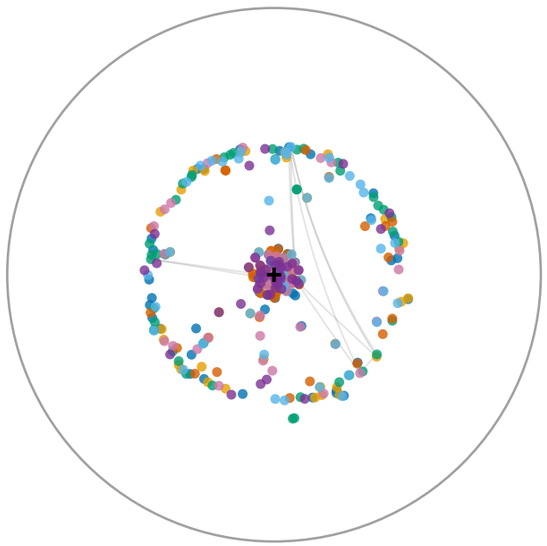
128D · actual radius · best epoch 7
Strict cone satisfaction: 0.00%



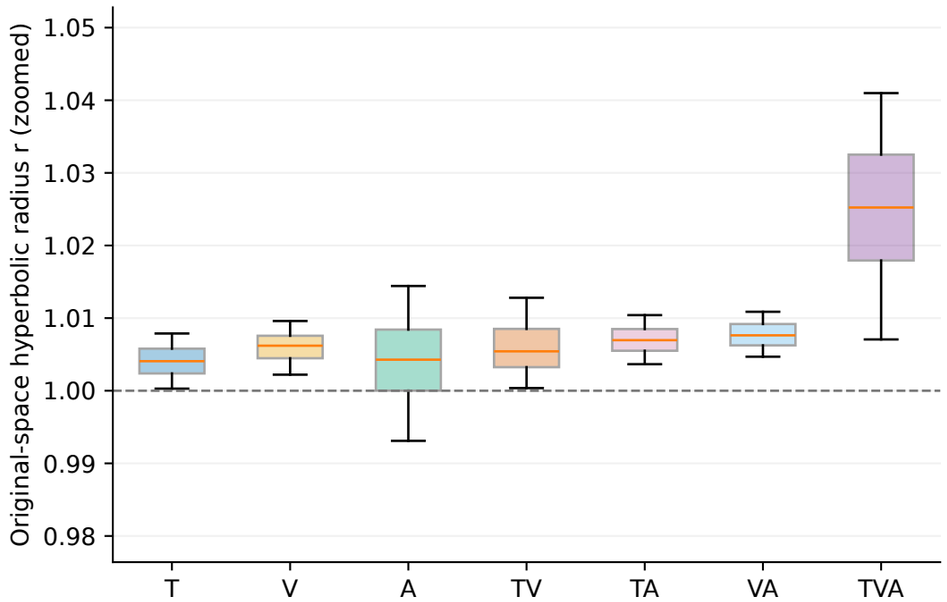
512D · HoroPCA · projected radii are distorted
distance ρ=0.35; kNN10=24.8%



512D · CO-SNE · projected radii are distorted
distance ρ=0.06; kNN10=16.0%



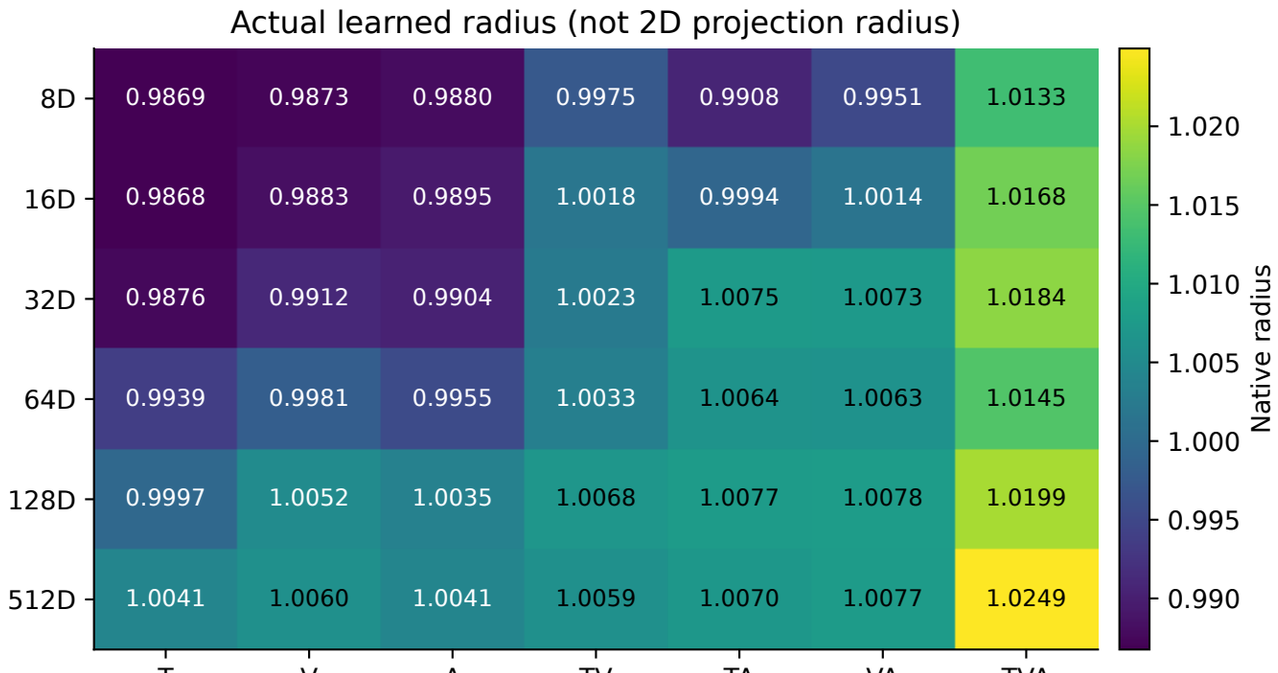
512D · actual radius · best epoch 5
Strict cone satisfaction: 0.00%

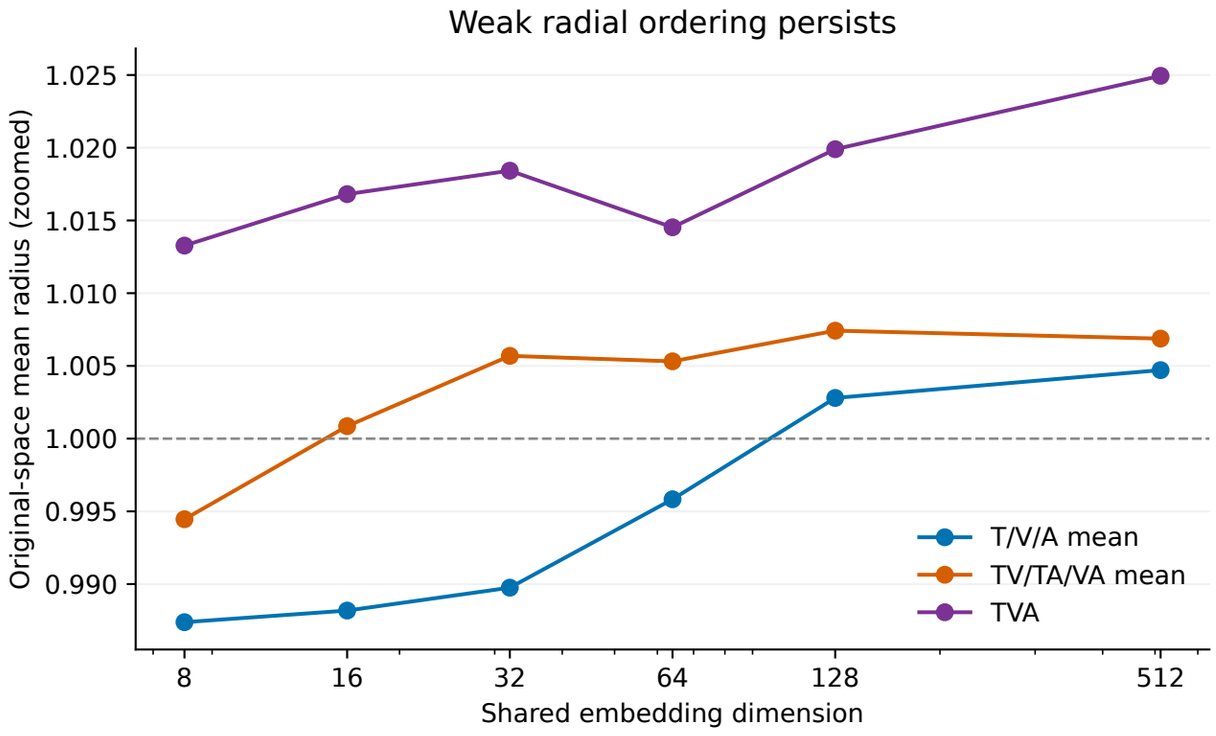


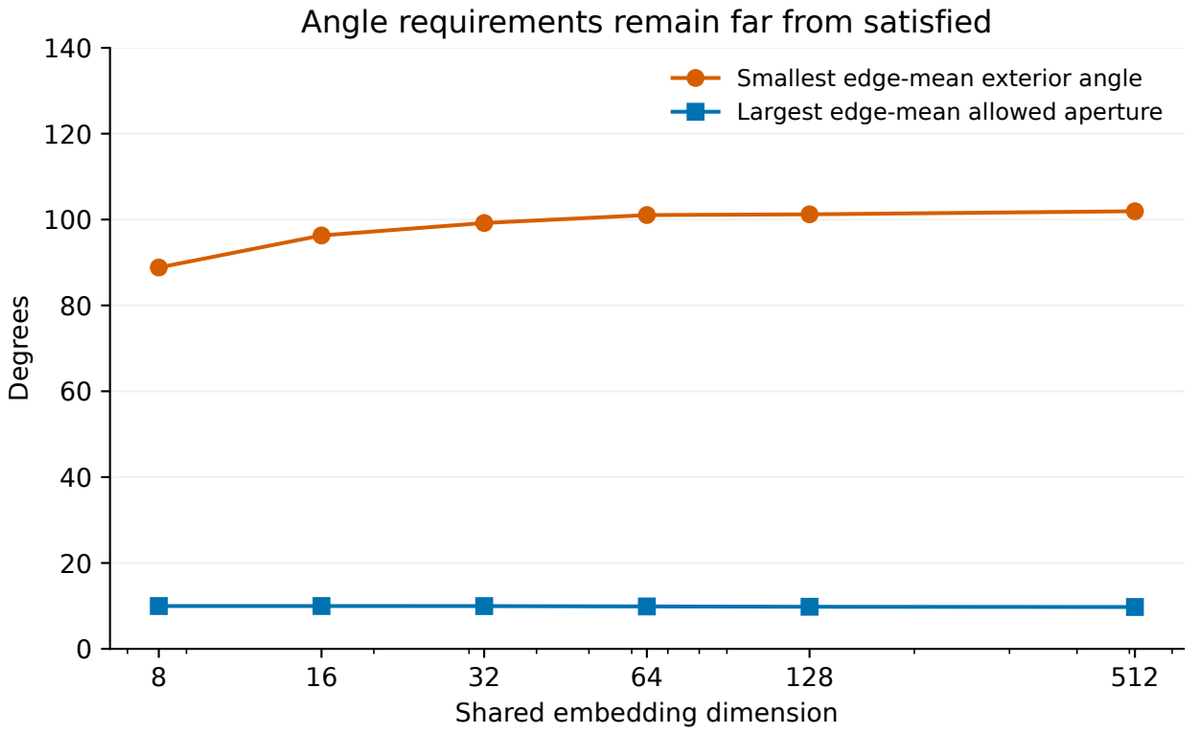
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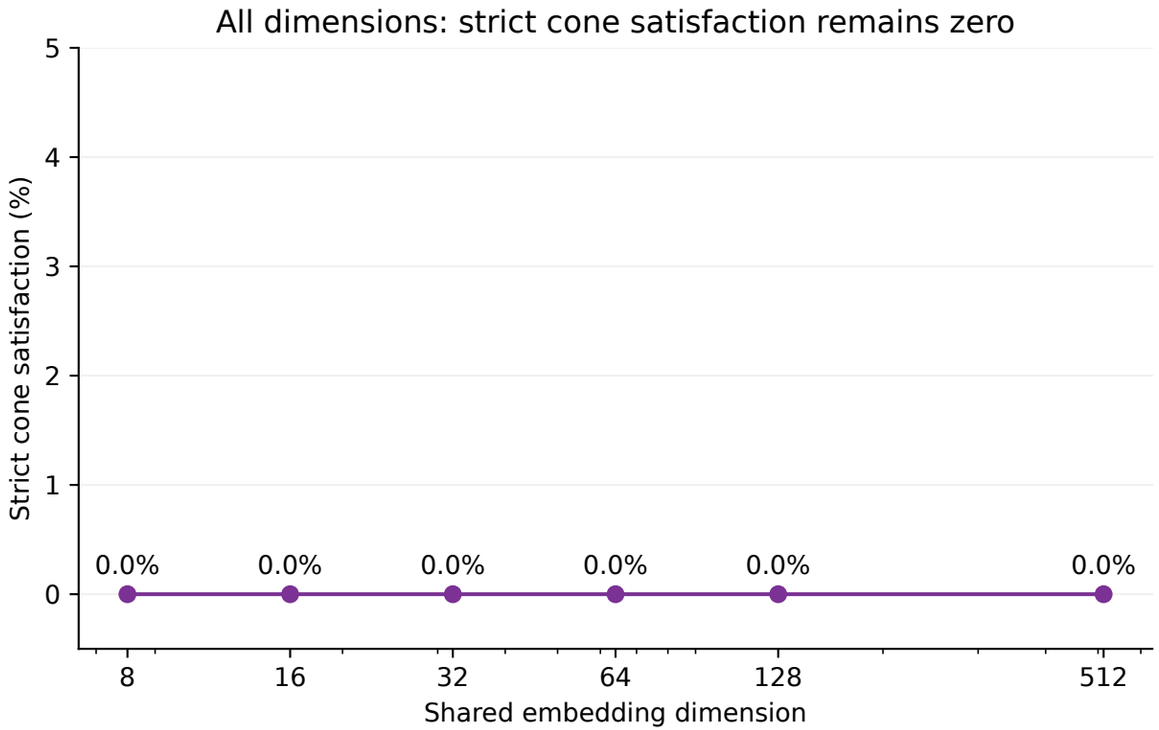
Same 48 test videos × 7 subsets; projection seed 50 only. Right column: all 884 videos. Gray lines are prescribed edges, not confirmed containment.

Native-space evidence · full test set · best checkpoints · lambda = 0.1

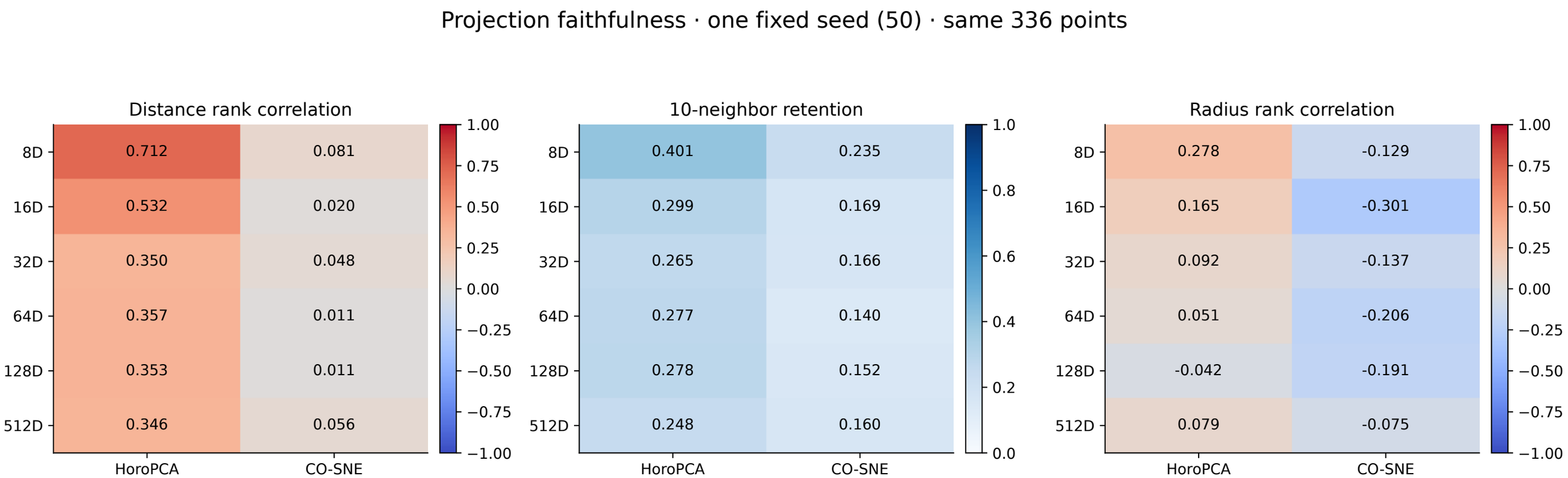








These are measured original-space radii and angles. The new lambda = 1.0 experiment is separate and is not included here.


Projection faithfulness · one fixed seed (50) · same 336 points
Distance rank correlation
8D
16D
32D
64D
128D
512D
0.712
0.081
0.532
0.020
0.350
0.048
0.357
0.011
0.353
0.011
0.346
0.056
HoroPCA
CO-SNE
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
10-neighbor retention
0.401
0.235
0.299
0.169
0.265
0.166
0.277
0.140
0.278
0.152
0.248
0.160
1.0
0.8
0.6
0.4
0.2
0.0
Radius rank correlation
0.278
-0.129
0.165
-0.301
0.092
-0.137
0.051
-0.206
-0.042
-0.191
0.079
-0.075
Low or negative radius correlation means a central TVA cluster in 2D cannot be interpreted as a small learned radius.